

Archive It with Zenodo

DOIs for your data, materials, and code (with GitHub releases)

Erin M. Buchanan

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Plan for Today

- Why archive, and what Zenodo is
- Key ideas: records, DOIs, versions, access
- Route 1: upload files directly (demo)
- Route 2: archive a GitHub release (demo)
- Tips for big teams
- Q&A, plus help with your own projects

Why Archive?

- Get credit for everything you make: data, materials, code, and protocols, not just papers
- Funders and journals increasingly ask for persistent identifiers (DOIs)
- OSF has changed, so many networks need a new home for files and DOIs
- Links break: websites and repos get renamed, moved, or deleted, but a DOI keeps working

What Is Zenodo?

- A general-purpose repository run by CERN
- Free to use, and the software is open source (InvenioRDM)
- Accepts any research output: data, code, slides, posters, protocols, preprints
- Up to 50 GB and 100 files per record
- Every record gets a DOI
- Log in at zenodo.org with ORCID, GitHub, or an email account

Key Idea: Records and Versions



- Each record is one thing you archive. Each version gets its own DOI.
- The concept DOI ties all the versions together.
- Need a change? Make a new version, not a new record.

See it in action: [jaspSyntheticData](#) has v0.1.0 and v0.2.0 under one concept DOI.

Key Idea: Locked Files, Editable Descriptions

Files: locked

After you publish, files can't be added, removed, or swapped. New files = new version.

Description: editable

Fix the title, authors, description, or links any time, with no new version needed.

Why that's good:

- Preregistrations and protocols get a permanent timestamp
- Every version stays exactly as it was, so changes are traceable
- The description grows with the project: add the paper, data, and code links later

Key Idea: Access Options

Choose who can download the files:

Open

Anyone can download the files

Restricted

People request access and you approve

Embargoed

Files open up automatically on a date you pick

The description is always public, even when the files aren't.

Practice First: The Sandbox

- sandbox.zenodo.org is a full copy of Zenodo just for testing
- Separate account; nothing there is permanent
- DOIs from the sandbox are not real

Route 1: Upload Directly

Upload: Basic Information

1. Go to My dashboard (top menu), then click New upload
2. Files: drag in your files
3. Basic information:
 - Digital Object Identifier: “Do you already have a DOI?” → No
 - Resource type: dataset, software, presentation...
 - Title and publication date
 - Creators: everyone in the citation, with ORCIDs
 - Description: what’s in the files and how to use them
 - Licenses: CC BY 4.0 by default. Help choosing: [Creative Commons chooser](#) or [choosealicense.com](#) for code

Upload: Recommended Info & Related Works

4. Recommended information:

- Contributors: helped, but not in the citation. Each needs a role (e.g., data collector)
- Keywords and subjects: your own, or from EuroSciVoc or MeSH
- Languages, version, and publisher

5. Related works: link the paper, preprint, preregistration, or repo

- Pick a relation (e.g., “Is supplement to”) and paste the DOI or URL

6. Choose access (open, restricted, embargoed), then Publish

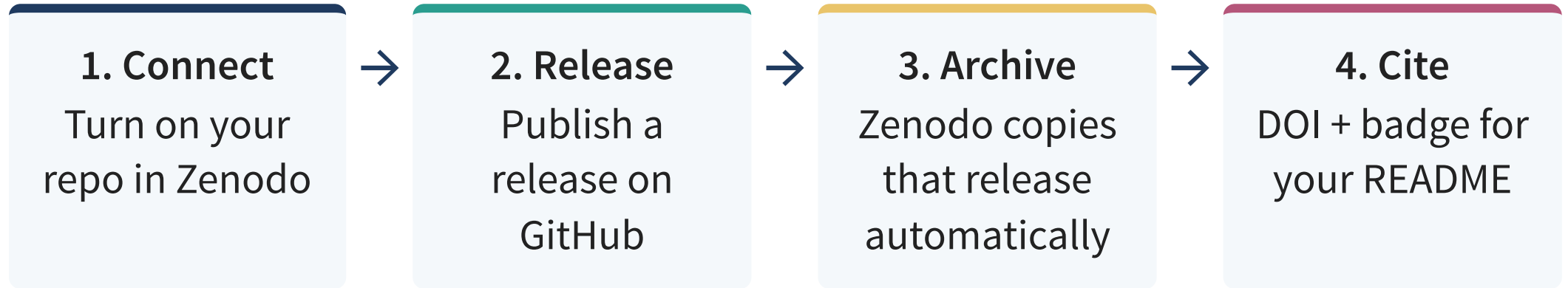
Live Demo: Upload

TODO: demo files to upload, e.g., a study protocol PDF and a data dictionary

1. Log in to the sandbox
2. Go to My dashboard → New upload, and add the files
3. Add creators with ORCIDs
4. Publish and look at the DOI
5. Make a new version of the record

Route 2: Archive a GitHub Release

How It Works



Every new GitHub release becomes a new version of the same Zenodo record.

Connecting GitHub, Step by Step

In Zenodo:

1. Log in, then open your profile menu → GitHub
2. Click Sync now
3. Find your repository and flip the switch to on

In GitHub:

4. Go to your repo → Releases → Draft a new release
5. Add a tag (e.g., **v1.0.0**), a title, and notes
6. Click Publish release

Back in Zenodo: wait a few minutes, then find your new DOI on the GitHub page.

Describe It: CITATION.cff

Add a `CITATION.cff` file to your repo so Zenodo (and GitHub) know how to cite it:

```
1 cff-version: 1.2.0
2 title: "Example BTS Project: Shared Analysis Code"
3 version: 1.0.0
4 license: "MIT"
5 type: software
6 message: "If you use this, please cite it as below."
7 authors:
8   - given-names: Erin M.
9     family-names: Buchanan
10    affiliation: Harrisburg University
11    orcid: "https://orcid.org/0000-0002-9689-4189"
12   - given-names: "[Next]"
13     family-names: "[Author]"
14 keywords:
15   - big team science
```

Bonus: GitHub shows a “Cite this repository” button.

Or: .zenodo.json

More control: contributors, funding, communities, and links to your paper. If both files exist, Zenodo uses only `.zenodo.json`.

```
1 {
2   "title": "Example BTS Project: Shared Analysis Code",
3   "upload_type": "software",
4   "license": "mit",
5   "creators": [
6     {"name": "Buchanan, Erin M.",
7       "affiliation": "Harrisburg University",
8       "orcid": "0000-0002-9689-4189"}
9   ],
10  "contributors": [
11    {"name": "Site Lead, Example", "type": "DataCollector"}
12  ],
13  "related_identifiers": [
14    {"identifier": "https://doi.org/[paper DOI]",
15     "relation": "isSupplementTo",
16     "resource_type": "publication-article"}
17  ],
18  "communities": [{"identifier": "[your-network]"}]
19 }
```

Live Demo: GitHub Release

github.com/doomlab/btscon-2026: the slides from this conference!

1. Switch the repo on in the Zenodo sandbox
2. Show `CITATION.cff` and `.zenodo.json`. This repo has both, so Zenodo uses `.zenodo.json`
3. Publish release `v1.0.0` on GitHub
4. Find the new record and DOI in Zenodo
5. Add the DOI badge to the README
6. Publish `v1.1.0` and watch a new version appear

When It Doesn't Work

- No record appeared? Check the repo was switched on *before* the release, and that the release was published (not a draft).
- Release failed? In Zenodo's GitHub page, click the release, then Errors. It's usually the metadata.
- Bad `CITATION.cff`? Validate it with `cffconvert` first.
- Wrong authors or title? Fix the metadata file, then publish a new release.

Tips for Big Teams

Zenodo for Big Team Science

- Make a Zenodo Community for your network to collect every project's outputs in one place
- Get everyone's ORCID before you publish, because author lists are long
- Link everything with related identifiers: preprint ↔ data ↔ code ↔ materials
- One record per output, with new versions when things change
- Use restricted access for sensitive data so it's citable but controlled
- Add your funder (e.g., SSHRC grant) in the metadata

Cheat Sheet

Zenodo	zenodo.org
Practice	sandbox.zenodo.org
Help docs	help.zenodo.org
GitHub integration	help.zenodo.org/docs/github
Make a CITATION.cff	cff-initializer
GitHub releases refresher	Pre-conference workshop recording

Questions?

Let's archive something!

Bring your own repo or files and we'll work through them together.